

MINI PROJECT

BY KANITHRA R

Project Title

Smart Waste Management System

A Real-Life Software Solution for Waste
Management

TITLE SLIDE:

SMART WASTE MANAGEMENT SYSTEM

A Real-Life Software Solution for Waste Management

Submitted By

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INTRODUCTION:

- Waste management is one of the biggest environmental challenges faced by cities and villages. Improper disposal of waste causes pollution, health issues, unpleasant surroundings, and environmental damage.
- Traditional waste collection methods rely on fixed schedules, resulting in overflowing bins in some locations while other bins are collected unnecessarily. This increases fuel consumption, labor costs, and pollution.
- The Smart Waste Management System is a software-based solution that uses digital technology to monitor waste collection, optimize routes, and improve communication between citizens and municipal authorities. The system helps create cleaner and healthier communities.

PROBLEM STATEMENT:

- Many cities face several waste management issues such as:
- Overflowing garbage bins
- Delayed waste collection
- Lack of proper waste segregation
- Manual monitoring of bins
- Increased operational costs
- Poor communication between citizens and authorities
- Environmental pollution
- Spread of diseases due to accumulated waste
- These problems highlight the need for a smart and automated waste management system.

EXISTING PROBLEM:

The existing waste management system operates manually.

- Features
 - Fixed collection schedule
 - Manual inspection of bins
 - Paper-based record maintenance
 - No real-time monitoring
 - No route optimization
- Limitations
 - Overflowing bins
 - Fuel wastage
 - High manpower requirement
 - Increased operational cost

PROPOSED MODEL:

The proposed system is a web-based software solution that digitizes waste management.

- Features
- Online waste monitoring
- Smart bin status tracking
- User complaint management
- Collection scheduling
- Route optimization
- Real-time notifications
- Waste segregation management
- Data analytics dashboard
- Report generation

OBJECTIVES:

- Improve waste collection efficiency.
- Reduce overflowing garbage bins.
- Minimize fuel consumption.
- Enable real-time monitoring.
- Encourage waste segregation.
- Provide an online complaint portal.
- Generate automated reports.
- Support clean and green initiatives.
- Improve public health.
- Promote sustainable waste management.

SYSTEM ARCHITECTURE:

Citizen

Frontend (React/HTML/CSS)

Backend (Node.js + Express)

MySQL Database

Admin Dashboard

Waste Collection Team

FRONTEND:

1. Home Page (index.html)

Features:

- Navigation Bar
- Project Introduction
- Login Button
- Register Button
- About Waste Management

Example sections:

FRONTEND: HTML

- Smart Waste Management System
- Home
- About
- Services
- Login
- Register
- Welcome to Smart Waste Management
- Manage waste efficiently using technology.
- [Login]
- [Register]

FRONTEND:

2. Login Page

- Fields
- Email
- Password
- Buttons
- Login
- Forgot Password

3. Registration Page

- Fields
- Name
- Email
- Phone
- Address
- Password
- Confirm Password
- Button
- Register

FRONTEND:

- Total Bins
- Complaints
- Collected Waste
- Pending Complaints
- Completed Collection

Reports
Display
Daily Collection

Monthly Collection

Most Filled Bins

Pending Complaints

BACKEND:

- This backend provides a clean, modular starting point suitable for a mini-project. For a production-ready application, you would typically add password hashing (e.g., bcrypt), authentication (JWT), input validation, proper error handling, logging, and environment variables for database credentials.

PROJECT MODULES:

Module 1 – User Management

- User Registration
- User Login
- Profile Management

Module 2 – Waste Bin Management

- Add Bin
- Update Bin
- Delete Bin
- View Bin Status

Module 3 – Complaint Management

- Register Complaint
- Upload Photo
- Track Complaint Status

PROJECT MODULES:

- Module 4 – Collection Management
 - Vehicle Assignment
 - Collection Scheduling
 - Collection Status Updates
- Module 5 – Reports & Analytics
 - Daily Reports
 - Monthly Reports
 - Waste Collection Statistics
 - Dashboard Analytics

SOFTWARE REQUIREMENTS:

- Frontend
 - HTML5
 - CSS3
 - JavaScript
 - React.js
- Backend
 - Node.js
 - Express.js
- Database
 - MySQL
- Tools
 - VS Code
 - Git
 - GitHub
 - Postman
 - XAMPP
- Operating System
 - Windows 10/11

WORKFLOW:

User Login

Locate Waste Bin

Report Complaint

Admin Receives Request

Assign Collection Vehicle

Waste Collection

Update Status

Generate Reports

ADVANTAGES:

- Real-time monitoring
- Reduced pollution
- Faster complaint resolution
- Improved public hygiene
- Fuel saving
- Reduced manpower
- Better resource utilization
- Digital record maintenance
- Data-driven decisions
- Eco-friendly solution
- Time saving
- Cost effective

REAL LIFE ACTIVITIES:

- The system can be used in:
- Smart Cities
- Municipal Corporations
- Town Panchayats
- Villages
- Colleges
- Universities
- Hospitals
- Shopping Malls
- Railway Stations
- Airports
- Hotels
- Residential Apartments
- Industrial Areas

CONCLUSION:

- The Smart Waste Management System is a practical software solution designed to improve waste collection and management. It replaces manual processes with a digital platform that enables efficient monitoring, complaint handling, route planning, and reporting. By reducing operational costs, improving cleanliness, and promoting recycling, the system contributes to environmental sustainability and supports the development of smart cities. Future integration with IoT, AI, and cloud technologies can make the solution even more intelligent and scalable.

THANK YOU

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